Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2025	ENGI3491-DE01

Title: (BEng/MEng)

Digital Electronics And Digital Signal Processing

Time Allowed:	2 hours
Additional Material provided:	No
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	This is a synchronous, time restricted exam. All submissions must be uploaded to Gradescope as per the published timetable. Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the exam submission window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.* • *In line with the Department of Engineering document entitled "Student Guidance on the Use of Generative Artificial Intelligence and Related Technology for the Department of Engineering" policy in assessments, the use of generative artificial intelligence tools and related technologies is strictly prohibited for this exam.*

Revision:	

Question 1

a) During the technical interview for a position in embedded-system design, the hiring manager asks the candidate to write a VHDL code for a full adder, based on these two logic functions:

```
cout = (a and b) or (cin and (a or b))
s = ((not cout) and (a or b or cin)) or (a and b and cin)
```

where `a` and `b` are the input bits to be added, `cin` is the input carry, `s` is the sum, and `cout` is the output carry.

The solution provided by the candidate is the following VHDL code:

```
library IEEE;
use IEEE.std_logic_1164.all;

entity fa is
port (a,b,cin : in std_logic;
      s, cout : out std_logic);
end fa;

architecture fa_arch of fa is
  signal cout_temp: std_logic;
begin
  main_proc: process(a,b,cin)
  begin
    cout_temp <= (a and b) or (cin and (a or b));
    cout <= cout_temp;
    s <= ((not cout_temp) and (a or b or cin)) or (a and b and cin);
  end process main_proc;
end fa_arch;
```

Unfortunately (for the candidate), this code does not produce the expected behaviour.

i) Explain how the code behaves and why the result is wrong, providing a few input cases.

ii) Alter the code to achieve the desired behaviour using the same logic functions.

[40%]

b) You have been tasked with the design in VHDL of an 8-bit adder. The design of a 4-bit adder already exists, and its top entity is:

```
entity fourbitadder is
port (a : in  std_logic_vector(3 down to 0);
      b : in  std_logic_vector(3 down to 0);
    cin : in  std_logic;
      s : out std_logic_vector(3 down to 0);
   cout : out std_logic);
end fourbitadder;
```

Write the VHDL code for the 8-bit adder with the following specifications:

- Use a multiplexer-based architecture to avoid introducing further delay in the carry propagation.
- All the inputs are connected to the adder through D-type flip flops, which are triggered on the rising edge of a common clock.
- The adder has an asynchronous reset (active high) which turns the output sum and output carry to 0.
- Only use the std_logic library (i.e., what is provided in "`library IEEE; use IEEE.std_logic_1164.all;`").

Explain how your code works, with a clear motivation for each section. You can use VHDL comments and/or free text.

[60%]

Question 2

a) Design an iterative logic network to detect the presence of at least two non-adjacent ‘1’s in a 6-bit word. For example, both 100001 and 110101 contain at least two non-adjacent ‘1’s whereas 110000 does not. Your answer should include the simplified Boolean expressions necessary to implement the basic cell and any output logic. State any assumptions and clearly show your working.

[30%]

b) A factory-based quality control system is used to check the mass, thickness and reflectivity of metal plates. The plates are on a conveyor belt and pass under appropriate sensors. The speed of the conveyor belt fluctuates throughout the day depending on the electrical power consumption elsewhere in the factory.

The sensors are all connected to a single sensor measurement module which provides three digital outputs, UVW. These outputs have the following meaning.

	UVW
Mass Pass	001
Mass Fail	101
Thickness Pass	011
Thickness Fail	111
Reflectivity Pass	010
Reflectivity Fail	110
Awaiting mass test	000

At the start, the quality control system has an overall output, XY = 00 and the sensor measurement module outputs, UVW = 000. First the mass is tested; then the thickness and then the reflectivity. If all three quality checks are passed then the system has outputs XY = 01. If any of the checks are failed then the outputs XY = 10.

continued

Once a test has been passed, it is not possible for a fail of the same test to be issued afterwards. It is also not possible for the sensor measurement module to output UVW = 000 after the first successful measurement of mass (unless it is reset by an operator). After the three tests, or if there is a fail, the output remains unchanged (again, unless the system is reset by an operator).

i) Design an asynchronous sequential logic network to meet the above specification.

   Your answer should include a state diagram, flow table(s) and excitation maps and all of the necessary Boolean expressions that would be required to implement the system. You should also clearly state any assumptions.

   You are not required to draw a circuit diagram. You do not need to include the reset behaviour in your design and you do not need to fully simplify the Boolean expressions.

ii) Describe how the provided specification helped support Fundamental Mode operation in your proposed implementation. Also, describe where it did not support Fundamental Mode operation and any impact this might have on realistic circuit implementations.

iii) Identify two benefits of using an asynchronous sequential logic design versus a synchronous solution for this quality control system implementation. Explain your reasoning.

[70%]

END